

Loss Prevention Recommendation Agent

保险损失预防推荐智能体 - 技术教程

Bilingual technical tutorial

Predictive risk / Uplift CATE / Causal forests / Agent tooling

双语技术教程：风险预测 / 增益建模 / 因果森林 / 智能体工具

Synthetic data only / 仅使用合成数据

Portfolio / research engineering project

面向技术面试与研究工程展示

0. Terminology Glossary

0. 术语表

This glossary is the source of truth for bilingual terminology used throughout the tutorial.
本术语表是全文中英术语一致性的基准。

English	中文
Loss Prevention	损失预防
Predictive Risk Model	预测风险模型
Uplift Modeling	增益建模 / 提升模型
CATE (Conditional Average Treatment Effect)	条件平均处理效应 (CATE)
Meta-learner (S/T/X)	元学习器 (S/T/X)
Causal Forest	因果森林
Confounding / Selection Bias	混杂 / 选择偏差
Propensity Score	倾向得分
Intervention Ranking	干预排序
Customer Burden	客户负担
Feasibility	可行性
Constraint Filtering	约束过滤
Counterfactual Simulation	反事实模拟
Agent Tool Calling	智能体工具调用
Explainability	可解释性
Calibration / ECE	校准 / 期望校准误差
Qini / AUUC	Qini 系数 / 增益曲线面积
Policy Value	策略价值
Explore vs Exploit	探索与利用
Responsible AI	负责任的人工智能

1. Project Overview

1. 项目概览

This project answers: given a customer/policy/property context, which intervention is most likely to reduce future insurance loss?

本项目回答：在给定客户/保单/财产上下文时，哪一种干预最有可能降低未来保险损失？

It is a research-engineering system demonstrating predictive ML, uplift/causal inference, recommender-style ranking, tool-using agents, explainability, and synthetic experimentation.

它是一个研究工程系统，展示预测机器学习、增益/因果推断、推荐式排序、工具型智能体、可解释性与合成实验。

Key capabilities

关键能力

- Calibrated P(future loss) risk scoring
校准后的未来损失概率风险评分
- Per-intervention CATE via S/T/X-learners and causal forest
基于 S/T/X 学习器与因果森林的逐干预 CATE
- Constraint-aware multi-objective ranking
带约束的多目标干预排序
- Grounded agent explanations (no invented treatment effects)
有依据的智能体解释（不编造处理效应）
- Interactive counterfactual simulator and research lab UI
交互式反事实模拟器与研究实验界面

KEY IDEA / 关键思想

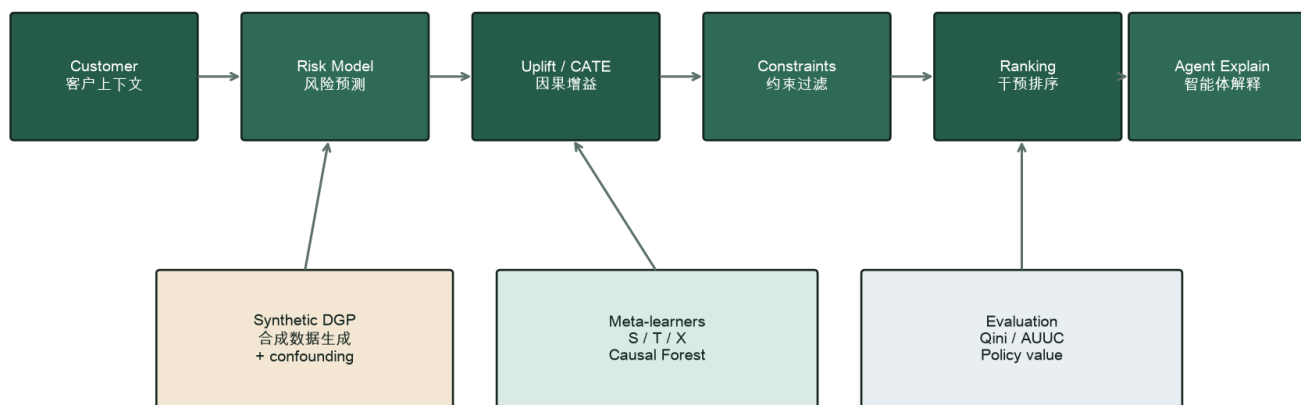
Predicting who is at risk is not the same as predicting who will benefit from an intervention.

预测谁有风险，并不等于预测谁会从干预中受益。

2. System Architecture

2. 系统架构

System Architecture / 系统架构



Online path: Customer → Risk Model → Risk Factors → Candidate Interventions → Uplift/Causal Model → Expected Impact → Constraint Filtering → Ranking → AI Explanation.

在线路径：客户 → 风险模型 → 风险因子 → 候选干预 → 增益/因果模型 → 期望影响 → 约束过滤 → 排序 → AI 解释。

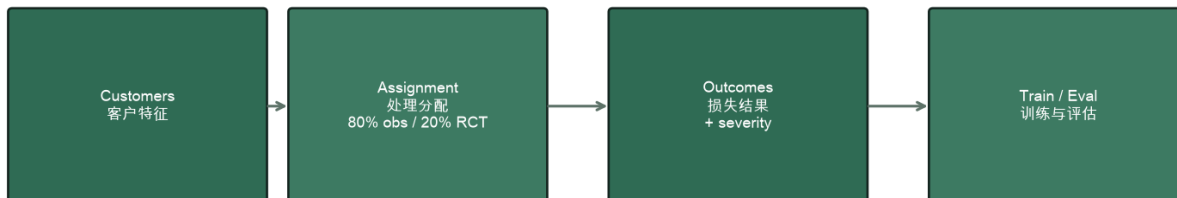
Implemented as a Python package (`'loss_prevention'`) with FastAPI service, static production UI, Typer CLI, and offline training/evaluation modules. There is no external database; state is synthetic and in-memory/model-artifact based.

实现为 Python 包 (`'loss_prevention'`)，包含 FastAPI 服务、静态生产级 UI、Typer CLI，以及离线训练/评估模块。无外部数据库；状态基于合成数据与内存/模型产物。

3. Data Architecture

3. 数据架构

Synthetic Experiment Pipeline / 合成实验流水线



All data are synthetic. The DGP creates customer features (maintenance, hazard, claims history, engagement, contactability, etc.) and historical intervention assignments.

全部数据为合成数据。数据生成过程创建客户特征（维护、危险暴露、理赔历史、参与度、可联系性等）与历史干预分配。

80% of assignments are observational with confounding: ops preferentially treats higher-risk and more contactable customers. 20% are RCT arms with propensity 0.5 for unconfounded reference.

80% 分配为带混杂的观测数据：运营更倾向处理高风险、更易联系的客户。20% 为倾向得分=0.5的随机对照试验臂，用作无混杂参照。

Why naive ATE misleads

为何朴素 ATE 会误导

Because treated and control groups differ systematically in baseline hazard, a naive control-minus-treated loss-rate comparison mixes selection differences with true effects.

由于处理组与对照组在基线危险上系统性不同，朴素的“对照损失率-处理损失率”会把选择差异与真实效应混在一起。

4. Machine Learning

4. 机器学习

Predictive model

预测模型

$$\mu(X) = P(Y=1 | X)$$

Implemented as gradient boosting with isotonic calibration. Trained preferentially on untreated rows to approximate baseline risk without treatment contamination.

实现为带等渗校准的梯度提升。优先在未处理样本上训练，以近似不受干预污染的基线风险。

Uplift / CATE

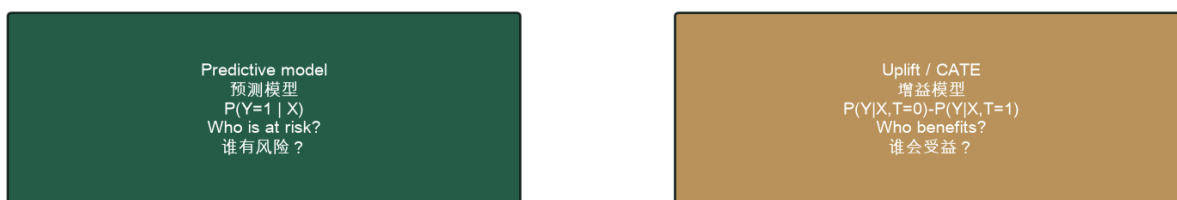
增益 / CATE

$$\tau(X, T) = P(Y=1 | X, T=0) - P(Y=1 | X, T=1)$$

Positive tau means absolute risk reduction. Meta-learners: S-learner (single model with T), T-learner (separate outcome models), X-learner (imputed effects + propensity weighting).

正的 tau 表示绝对风险下降。元学习器：S (含处理指示的单模型)、T (处理/对照分离模型)、X (效应填补 + 倾向加权)。

Prediction vs Treatment Effect / 预测 vs 干预效应



High risk customer may have low CATE for a given intervention
高风险客户对某干预的增益可能很低

Causal forest

因果森林

Educational honest forest: structure sample vs estimation sample; splits encourage treatment-effect heterogeneity; leaf CATE is control-minus-treated loss rate.

教学用途的诚实森林：结构样本与估计样本分离；分裂鼓励处理效应异质性；叶节点为对照与处理损失率之差。

CATE

Metrics

指标

- ROC-AUC, Brier, log loss, ECE for prediction/calibration

ROC-AUC、Brier、对数损失、ECE 用于预测/校准

- Qini and AUUC for uplift ranking quality
Qini 与 AUUC 用于增益排序质量
- Policy value = mean oracle CATE under a decision policy
策略价值 = 决策策略下 oracle CATE 均值

5. AI / LLM Architecture

5. AI / 大模型架构

The LLM is optional and disabled unless OPENAI_API_KEY is set. Deterministic model services own all numeric scores.

大模型为可选项，仅在设置 OPENAI_API_KEY 时启用。所有数值分数由确定性模型服务给出。

If enabled, the LLM may rewrite narrative text but is instructed not to invent probabilities, CATEs, costs, or confidence scores.

若启用，大模型可润色叙述，但被明确要求不得编造概率、CATE、成本或置信度。

KEY IDEA / 关键思想

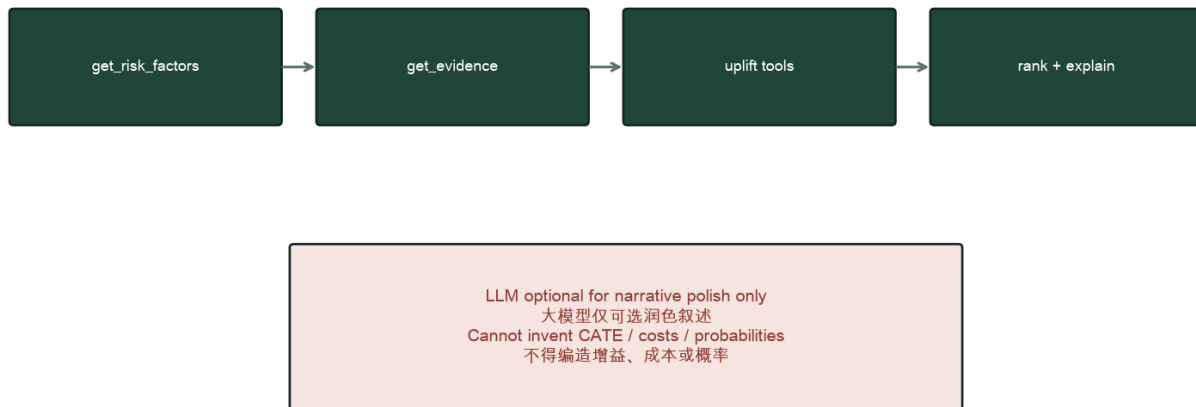
This separation is intentional: LLMs are good at communication; bad as silent inventors of causal effects.

这种分离是有意设计：大模型擅长表达，不适合默默编造因果效应。

6. Agent Architecture

6. 智能体架构

Agent Tool Flow / 智能体工具流



LossPreventionAgent orchestrates tools: inspect risk factors, retrieve evidence, score candidates, call uplift/causal tools, compare interventions, explain trade-offs, emit structured Recommendation.

LossPreventionAgent

编排工具：检查风险因子、检索证据、给候选打分、调用增益/因果工具、比较干预、解释权衡、输出结构化 Recommendation。

- get_risk_factors / get_evidence_summary
获取风险因子 / 证据摘要
- get_uplift_estimates / get_ranked_interventions
获取增益估计 / 排序结果
- simulate_intervention (counterfactual)
干预反事实模拟
- compare_interventions
比较两种干预

7. Ranking & Constraints

7. 排序与约束

```
utility = w_b*CATE + w_c*confidence + w_f*feasibility - w_cost*cost_norm - w_burden*burden
```

Hard filters: recent-contact suppression, annual intervention caps, fraud/specialist suitability, burden limits for low-engagement customers. no_intervention is first-class.

硬约束：近期联系抑制、年度干预上限、欺诈/专家适用性、低参与客户负担限制。无干预（no_intervention）是一等选项。

8. Engineering Architecture

8. 工程架构

Package layout: data/, models/, ranking/, agent/, explain/, evaluation/, api/, ui/static/. CLI entrypoint: loss-prevention. API serves both JSON and the production UI.

包结构 : data/、models/、ranking/、agent/、explain/、evaluation/、api/、ui/static/。CLI 入口 : loss-prevention。API 同时提供 JSON 与生产 UI。

Important API endpoints (implemented)

重要 API 端点 (已实现)

- GET /customers/{id}/recommend — recommendation + agent_trace
推荐结果 + 智能体轨迹
- POST /counterfactual — what-if simulator
反事实模拟
- GET /customers/{id}/method-compare — S/T/X/CF comparison
元学习器比较
- GET /research/deploy|experiments — offline research metrics
离线研究指标
- GET /architecture — pipeline/tool metadata for UI
架构元数据

9. Evaluation & Testing

9. 评估与测试

Automated tests cover synthetic DGP properties, predictive/uplift/causal models, ranking constraints, agent grounding, API contracts, static UI assets, and HTTP end-to-end journeys.

自动化测试覆盖合成数据性质、预测/增益/因果模型、排序约束、智能体证据落地、API 契约、静态 UI 资源，以及 HTTP 端到端旅程。

Research experiments investigate selection bias, treatment heterogeneity, policy optimization, and explore-vs-exploit diversity/value tradeoffs.

研究实验考察选择偏差、处理异质性、策略优化，以及探索-利用的多样性/价值权衡。

10. Design Decisions

10. 设计决策

Why uplift instead of risk-only targeting?

为何不用纯风险定向？

Risk-only policies over-spend on unresponsive high-risk customers. Uplift ranking optimizes expected risk reduction under cost/burden constraints.

纯风险策略会在无响应高风险客户上过度花费。增益排序在成本/负担约束下优化期望风险下降。

Why not let the LLM choose the intervention numerically?

为何不让大模型直接给出数值决策？

Causal effects require experimental/observational identification assumptions. LLMs do not estimate CATEs from first principles in this system; inventing them would be unsafe for a decision product.

因果效应需要识别假设。本系统中大模型不会凭空估计 CATE；编造数值对决策产品不安全。

Why synthetic data?

为何使用合成数据？

Insurance microdata are confidential. A transparent DGP with oracle CATEs enables honest evaluation of confounding and policy value.

保险微观数据涉密。带 oracle CATE 的透明数据生成过程，可诚实评估混杂与策略价值。

11. Limitations

11. 局限

- Synthetic DGP $\neq$ real-world claims processes
合成数据生成过程 $\neq$ 真实理赔过程
- Observational CATE estimates remain assumption-dependent
观测 CATE 估计仍依赖假设
- Causal forest is educational, not econml-production scale
因果森林为教学实现，非生产级 econml 规模
- No persistent customer database or auth layer
无持久客户库与鉴权层
- Optional LLM path not required for core recommendations
核心推荐不依赖可选大模型路径

12. Future Improvements

12. 未来改进

- Integrate real RCT logs / doubly robust estimators
接入真实 RCT 日志 / 双重稳健估计
- Online bandit allocation with delayed claim outcomes
带延迟理赔结果的在线老虎机分配
- Model registry, drift monitors, and canary policies
模型注册、漂移监控与金丝雀策略
- Stronger accessibility tooling and browser E2E (Playwright) in CI
更强无障碍工具与 CI 中的浏览器 E2E (Playwright)

13. How to Run

13. 运行方式

```
cd loss-prevention
source ../.venv/bin/activate
pip install -e . --no-deps
loss-prevention train
loss-prevention serve-ui
# open http://127.0.0.1:8010/
pytest -q
```

Open the UI to walk Risk Overview -> Interventions -> Counterfactual -> Model Compare -> Research Lab.

打开 UI，依次体验风险总览 -> 干预排序 -> 反事实 -> 模型比较 -> 研究实验。